

Module 6: Organic chemistry and analysis

The content within this module assumes knowledge and understanding of the chemical concepts developed in Module 2: Foundations in chemistry and Module 4: Core organic chemistry.

This module introduces several new functional groups and emphasises the importance of organic synthesis. This module also adds NMR spectroscopy to the instrumentation techniques used in organic and forensic analysis.

The main areas of organic chemistry studied include:

- aromatic compounds
- carboxylic acids and esters
- organic nitrogen compounds: amines and amino acids
- polymerisation: addition polymers and condensation polymers
- synthetic organic chemistry and further development of practical skills
- the importance of modern analytical techniques in organic analysis.

Synoptic assessment

This module provides a context for synoptic assessment and the subject content links strongly with the content encountered in Module 2: Foundations in chemistry and Module 4: Core organic chemistry.

- Atoms, moles and stoichiometry
- Acid and redox reactions
- Bonding and structure
- Organic nomenclature and structures
- Hydrocarbons
- Alcohols and haloalkanes
- Synthesis and analysis

Knowledge and understanding of Module 2 and Module 4 will be assumed and examination questions will be set that link their content with this module and other areas of chemistry.